

**Quick or slow?**

Bread dough made with yeast rises slowly through fermentation. In this process, chemical compounds in the yeast react with sugar to produce bubbles of carbon dioxide gas, which make the dough rise. With baking soda, the reaction is between an acid and an alkali, which generates carbon dioxide in an instant.

**Hot or cold?**

It takes energy to break the bonds between atoms, while energy is released when new bonds form. Often, more energy is released than it takes to break the bonds. That energy is released as heat, such as when a candle burns. This is an exothermic reaction. If the energy released is less than the energy required to break the bonds, the reaction takes energy from its surroundings and both become colder. That reaction is endothermic.

**Redox reactions**

Redox reactions involve reduction (the removal of oxygen, or addition of electrons) and oxidation (the addition of oxygen or removal of electrons). When an apple turns brown in the air, a chemical inside the apple is oxidized, and oxygen from the air is reduced.

**Reversible or irreversible?**

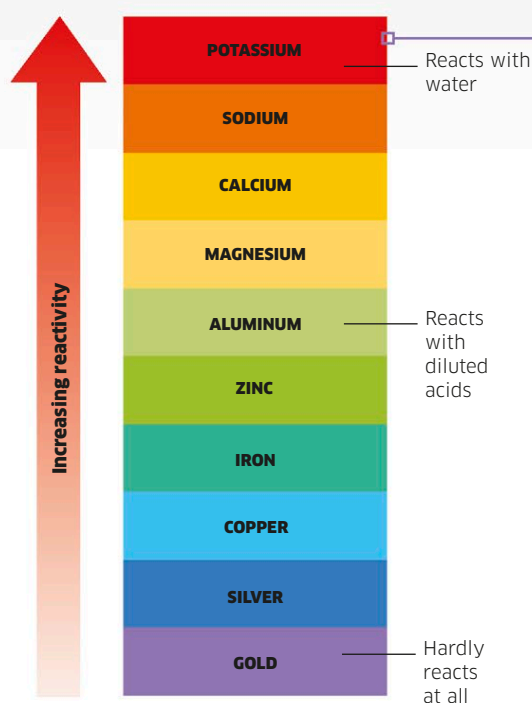
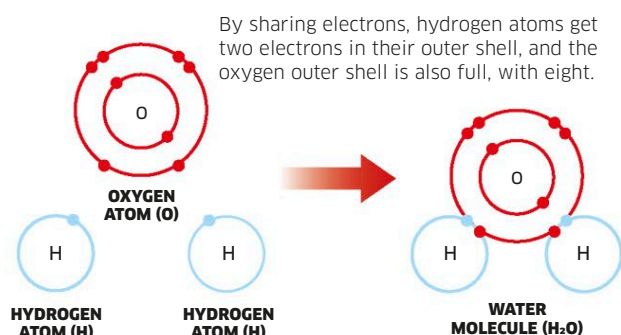
Rusting is a redox reaction that, like an apple going brown, is irreversible. In a reversible reaction, certain products can turn back into their original reactants.

**WHY DO REACTIONS HAPPEN?**

Different chemical reactions happen for different reasons, including temperature, pressure, and the type and concentration of reactants. Chemical reactions involve the breaking and making of bonds between atoms. These bonds involve the electrons in the outer shell of each atom. It is how the electrons are arranged in atoms of different elements that decides which atoms can lose electrons and which ones gain them.

**Why do atoms react?**

Atoms that can easily lose electrons are likely to react with atoms that need to fill their outer shell. There are different types of bonds depending on how the atoms do this: covalent, ionic, and metallic (see pp.16–17). A water molecule (below) has covalent bonds.

**Metal reactivity series**

A reactivity series sorts elements according to how readily they react with other elements. The most reactive is at the top; the least reactive at the bottom. It helps predict how elements will behave in some chemical reactions.

**Potassium**

Potassium is the most reactive metal in the series. Adding a lump of potassium to water causes the potassium to react instantly: it whizzes around on the surface of the water and bursts into spectacular flames.

